NAME: BENNY
SURNAME: ISSANZA

DAE INTEGRATED PROJECT PART 1 – ASSIGNMENT

Section1_Exploration

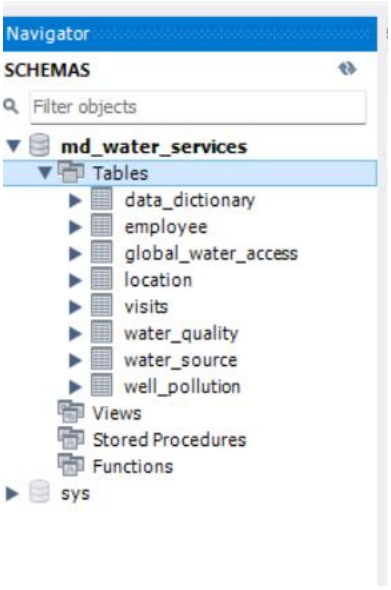
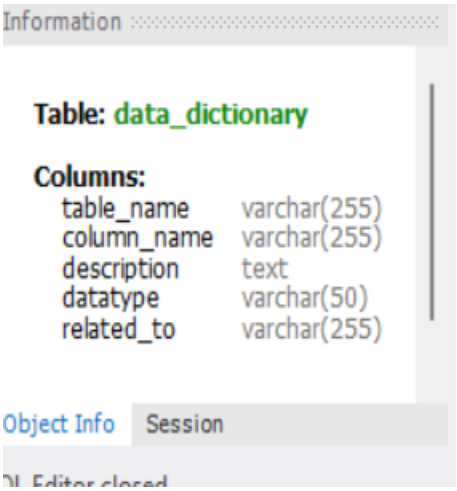
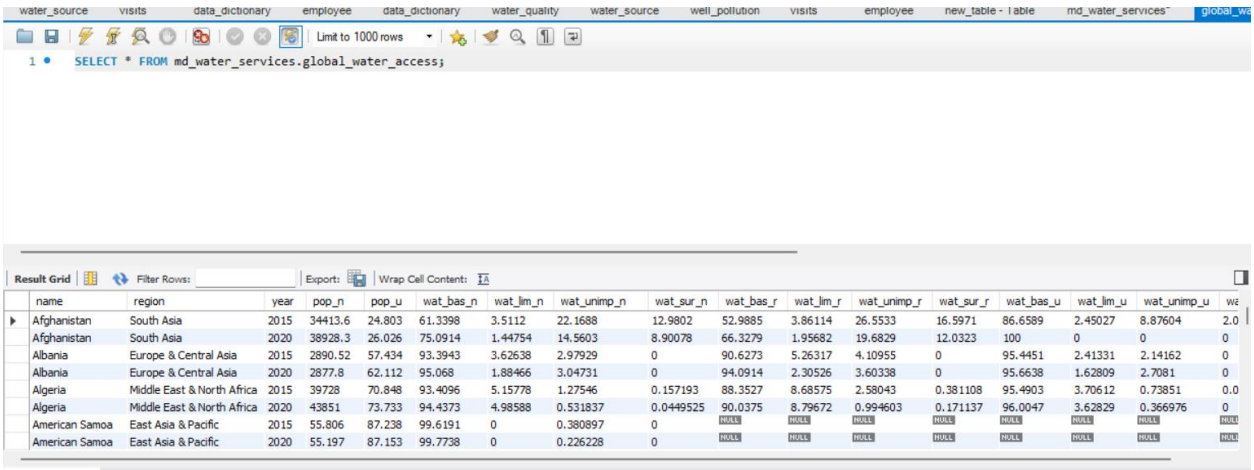


Table structure



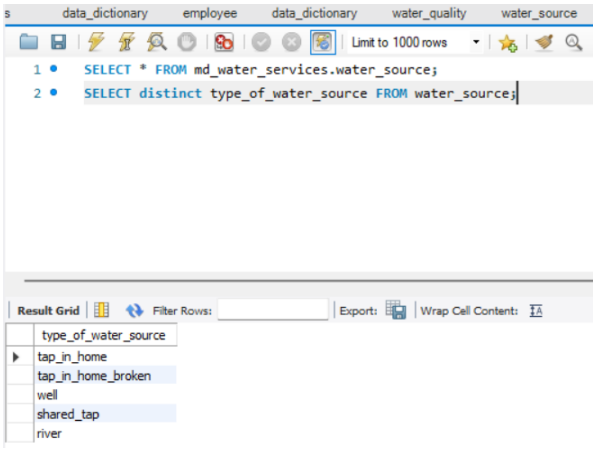
Column Names



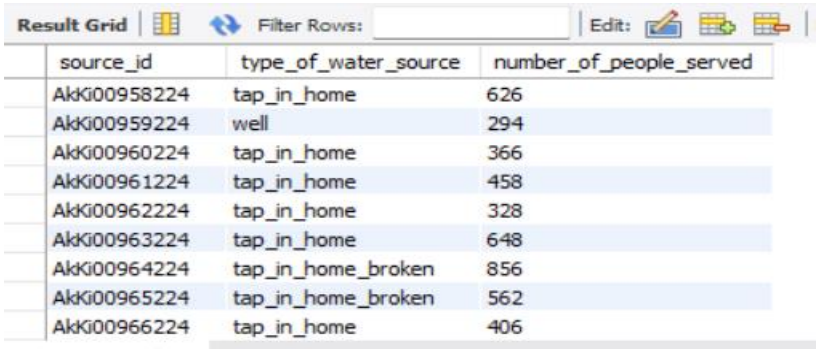
name	region	year	pop_n	pop_u	wat_bas_n	wat_lim_n	wat_unimp_n	wat_sur_n	wat_bas_r	wat_lim_r	wat_unimp_r	wat_sur_r	wat_bas_u	wat_lim_u	wat_unimp_u
Afghanistan	South Asia	2015	34413.6	24.803	61.3398	3.5112	22.1688	12.9802	52.9885	3.86114	26.5533	16.5971	86.6589	2.45027	8.87604
Afghanistan	South Asia	2020	38928.3	26.026	75.0914	1.44754	14.5603	8.90078	66.3279	1.95682	19.6829	12.0323	100	0	0
Albania	Europe & Central Asia	2015	2890.52	57.434	93.3943	3.62638	2.97929	0	90.6273	5.26317	4.10955	0	95.4451	2.41331	2.14162
Albania	Europe & Central Asia	2020	2877.8	62.112	95.068	1.88466	3.04731	0	94.0914	2.30526	3.60338	0	95.6638	1.62809	2.7081
Algeria	Middle East & North Africa	2015	39728	70.848	93.4096	5.15778	1.27546	0.157193	88.3527	8.68575	2.58043	0.381108	95.4903	3.70612	0.73851
Algeria	Middle East & North Africa	2020	43851	73.733	94.4373	4.98588	0.531837	0.0449525	90.0375	8.79672	0.994603	0.171137	96.0047	3.62829	0.366976
American Samoa	East Asia & Pacific	2015	55.806	87.238	99.6191	0	0.380897	0	NULL	NULL	NULL	NULL	NULL	NULL	NULL
American Samoa	East Asia & Pacific	2020	55.197	87.153	99.7738	0	0.226228	0	NULL	NULL	NULL	NULL	NULL	NULL	NULL

One explored Table.

Section2_WaterSources

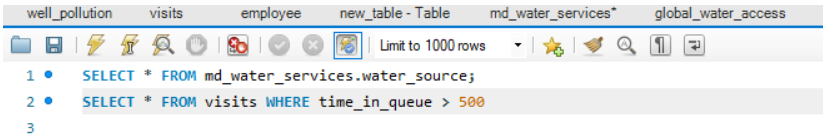


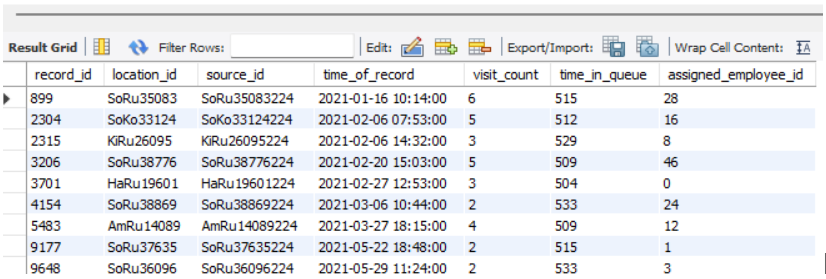
The Query and Different water types.



Number of people served.

Section3_Visits





Long que results.

```
SELECT * FROM md_water_services.water_source;
SELECT * FROM visits WHERE time_in_queue = 0
```

record_id	location_id	source_id	time_of_record	visit_count	time_in_queue	assigned_employee_id
1	KiRu28935	KiRu28935224	2021-01-01 09:17:00	1	0	46
3	AkLu01628	AkLu01628224	2021-01-01 09:53:00	1	0	1
6	AkRu05234	AkRu05234224	2021-01-01 10:18:00	1	0	30
7	KiRu28520	KiRu28520224	2021-01-01 10:28:00	1	0	34
8	HaZa21742	HaZa21742224	2021-01-01 10:37:00	1	0	6
9	AmDa12214	AmDa12214224	2021-01-01 10:58:00	1	0	36
10	HaRu19725	HaRu19725224	2021-01-01 11:04:00	1	0	16
13	AkKi00851	AkKi00851224	2021-01-01 11:14:00	1	0	32
14	AmDa11956	AmDa11956224	2021-01-01 11:16:00	1	0	6

Zero que results

Section4_Quality

```
SELECT * FROM md_water_services.water_quality;
SELECT * FROM water_quality WHERE subjective_quality_score =10 AND visit_count >1
```

record_id	subjective_quality_score	visit_count
5180	10	7
5181	10	5
5221	10	5
5262	10	6
5266	10	6
5279	10	2
5308	10	8
5310	10	7
5343	10	2

My query and filtered results

Section5_Pollution

```
SELECT * FROM md_water_services.well_pollution;
SELECT * FROM well_pollution where results='Clean' and biological>0.01
```

source_id	date	description	pollutant_ppm	biological	results
AkRu08936224	2021-01-08 09:22:00	Bacteria: E. coli	0.0406458	35.0068	Clean
AkRu06489224	2021-01-10 09:44:00	Clean Bacteria: Giardia Lamblia	0.0897904	38.467	Clean
SoRu38011224	2021-01-14 15:35:00	Bacteria: E. coli	0.0425095	19.2897	Clean
AkKi00955224	2021-01-22 12:47:00	Bacteria: E. coli	0.0812092	40.2273	Clean
KiHa22929224	2021-02-06 13:54:00	Bacteria: E. coli	0.0722537	18.4482	Clean
KiRu25473224	2021-02-07 15:51:00	Clean Bacteria: Giardia Lamblia	0.0630094	24.4536	Clean
HaRu17401224	2021-03-01 13:44:00	Clean Bacteria: Giardia Lamblia	0.0649209	25.8129	Clean
AkRu07137224	2021-03-04 13:41:00	Clean Bacteria: Giardia Lamblia	0.0656843	18.2978	Clean
KiRu27205224	2021-03-13 14:17:00	Clean Bacteria: Giardia Lamblia	0.0418018	49.4281	Clean

The incorrect records and contamination values together with the results.

Description errors

```
SELECT * FROM md_water_services.well_pollution;
SELECT * FROM well_pollution where description in ('Clean Bacteria: E. coli','Clean Bacteria: Giardia Lamblia')
```

source_id	date	description	pollutant_ppm	biological	results
AkRu06489224	2021-01-10 09:44:00	Clean Bacteria: Giardia Lamblia	0.0897904	38.467	Clean
KiRu25473224	2021-02-07 15:51:00	Clean Bacteria: Giardia Lamblia	0.0630094	24.4536	Clean
HaRu17401224	2021-03-01 13:44:00	Clean Bacteria: Giardia Lamblia	0.0649209	25.8129	Clean
AkRu07137224	2021-03-04 13:41:00	Clean Bacteria: Giardia Lamblia	0.0656843	18.2978	Clean
KiRu27205224	2021-03-13 14:17:00	Clean Bacteria: Giardia Lamblia	0.0418018	49.4281	Clean
AkHa00514224	2021-04-11 12:11:00	Clean Bacteria: Giardia Lamblia	0.0305404	22.0255	Clean
AmAm09776224	2021-05-23 11:28:00	Clean Bacteria: Giardia Lamblia	0.0963821	13.6574	Clean
SoIl32894224	2021-07-11 11:37:00	Clean Bacteria: Giardia Lamblia	0.0712408	5.44957	Clean
AkRu07366224	2021-07-23 11:19:00	Clean Bacteria: Giardia Lamblia	0.0969458	26.0308	Clean

Query and incorrect descriptions found.

Section6_DataCleaning

```
SELECT * FROM md_water_services.well_pollution;
CREATE TABLE md_water_services.well_pollution_copy AS (SELECT*FROM md_water_services.well_pollution);
UPDATE well_pollution_copy SET description= 'Bacteria: E. coli' WHERE description='Clean Bacteria: E. coli';
UPDATE well_pollution_copy SET description= 'Bacteria: Giardia Lamblia' WHERE description='Clean Bacteria: Giardia Lamblia';
UPDATE well_pollution_copy SET results= 'Contaminated: Biological' WHERE results= 'Clean' AND biological>0.01
```

Result Grid

source_id	date	description	pollutant_ppm	biological	results
KiRu28935224	2021-01-04 09:17:00	Bacteria: Giardia Lamblia	0	495.898	Contaminated: Biological
AkLu01628224	2021-01-04 09:53:00	Bacteria: E. coli	0	6.09608	Contaminated: Biological
HaZa21742224	2021-01-04 10:37:00	Inorganic contaminants: Zinc, Zinc, Lead, Cadmi...	2.715	0	Contaminated: Chemical
HaRu19725224	2021-01-04 11:04:00	Clean	0.0288593	0.0000956996	Clean
SoRu35703224	2021-01-04 11:29:00	Bacteria: E. coli	0	22.5009	Contaminated: Biological
AkHa00070224	2021-01-04 11:42:00	Inorganic contaminants: Cadmium	5.46739	0	Contaminated: Chemical
HaSe21346224	2021-01-04 11:52:00	Clean	0.0140376	0.0000898989	Clean
HaYa21468224	2021-01-04 12:03:00	Inorganic contaminants: Chromium, Barium, Chr...	6.05137	0	Contaminated: Chemical
SoRu36278224	2021-01-04 12:24:00	Parasite: Cryptosporidium	0	485.162	Contaminated: Biological

Backup Table creation, update statements, and update records.

```
SELECT * FROM md_water_services.well_pollution;
UPDATE well_pollution SET description= 'Bacteria: E. coli' WHERE description='Clean Bacteria: E. coli';
UPDATE well_pollution SET description= 'Bacteria: Giardia Lamblia' WHERE description='Clean Bacteria: Giardia Lamblia';
UPDATE well_pollution SET results= 'Contaminated: Biological' WHERE results= 'Clean' AND biological>0.01
```

Result Grid

source_id	date	description	pollutant_ppm	biological	results
KiRu28935224	2021-01-04 09:17:00	Bacteria: Giardia Lamblia	0	495.898	Contaminated: Biological
AkLu01628224	2021-01-04 09:53:00	Bacteria: E. coli	0	6.09608	Contaminated: Biological
HaZa21742224	2021-01-04 10:37:00	Inorganic contaminants: Zinc, Zinc, Lead, Cadmi...	2.715	0	Contaminated: Chemical
HaRu19725224	2021-01-04 11:04:00	Clean	0.0288593	0.0000956996	Clean
SoRu35703224	2021-01-04 11:29:00	Bacteria: E. coli	0	22.5009	Contaminated: Biological
AkHa00070224	2021-01-04 11:42:00	Inorganic contaminants: Cadmium	5.46739	0	Contaminated: Chemical
HaSe21346224	2021-01-04 11:52:00	Clean	0.0140376	0.0000898989	Clean
HaYa21468224	2021-01-04 12:03:00	Inorganic contaminants: Chromium, Barium, Chr...	6.05137	0	Contaminated: Chemical
SoRu36278224	2021-01-04 12:24:00	Parasite: Cryptosporidium	0	485.162	Contaminated: Biological

well_pollution 10

Verification queries and corrected results.